Econ490/690: Machine Learning for Finance

Details of Hedge Fund Exercise

This course centers around a semester-long experiential activity wherein you and a team of classmates will operate your own systematic, quantitatively focused hedge fund.

The purpose of this exercise is not simply to generate the highest realized investment return. Rather, the goal is to develop, implement, evaluate, communicate, and defend a systematic investment process that integrates finance, statistics, machine learning, and computation.

Throughout the semester, your fund should follow a repeated research cycle:

Develop an idea → Implement and test → Evaluate the evidence → Make a portfolio decision → Observe results → Revise

Not every research idea should become part of your live portfolio. A well-designed and correctly implemented analysis that leads your team to reject a proposed method may represent equally strong research as one that leads to deployment.

Project Structure

The exercise consists of three primary types of deliverables:

1. **Position Matrices**: Weekly submissions that translate your investment process into actual portfolio positions.
2. **Roadshows**: Approximately five cumulative presentations in which your team communicates and defends the evolution of its investment process to prospective investors.
3. **Due Diligence Materials**: A final cumulative package documenting and evaluating your fund's research process, portfolio performance, code, and investment decisions.

These deliverables are intended to build upon one another. In particular, Roadshows should represent updated versions of your fund's evolving investor presentation rather than five unrelated presentations created from scratch. Similarly, the final Due Diligence Materials should synthesize work conducted throughout the semester rather than constitute a new research project at the end of the course.

Fund Details

- **Trading**: 21Sep2026 through 2Dec2026
- **Fund Teams**: Each team consists of 2–4 people, depending upon class size.
- **Style**: Systematic, quantitatively focused, long/short.
- **Mandate**:
  - Primary: Absolute return.

  - Secondary: Risk-adjusted return relative to a benchmark. Sharpe Ratio will be the key risk-adjusted performance metric, although other measures may also be considered.

- **Academic Evaluation versus Investment Performance**:
  - Realized investment performance is an important outcome within the simulation, but it is not by itself a measure of the quality of your academic work.
  - High realized returns do not necessarily imply a high academic score, and poor realized returns do not necessarily imply a poor academic score.
  - Evaluation will emphasize the quality of your research process, implementation, evidence, reasoning, portfolio construction, and communication.

- **Trading Frequency**:
  - Strategies should generally be designed around a one-week holding period, beginning at the close of business on Friday and continuing through the close of business the following Friday.
  - Performance will be computed using daily adjusted closing prices.
  - Teams may alter positions within the week at End of Day (EOD), but frequent trading will be penalized as follows:

$$\widetilde{SR} = SR - \lambda \times TO,$$

    where TO is turnover during the week, computed as

$$TO_{week} = \sum_{t=Friday_1}^{Friday_2} \sum_i |w_{i,t} - w_{i,t-1}|,$$

    and $\lambda = 0.5$.
  - The purpose of the turnover adjustment is to discourage unnecessary trading and to encourage strategies that are consistent with the intended investment horizon.
  - The instructor reserves the right to alter the penalty if necessary to preserve the intended pedagogical purpose of the exercise.

- **Investable Universe and Benchmark**:
  - The investable universe covers stocks, Mutual Funds, and ETFs for which the required data are available through Yahoo Finance.
  - The benchmark is the S&P 500, as proxied by IVV.
  - Violations of the investable universe and/or benchmark rules will adversely affect the team's grade.

- **Investment Process**:
  - Since you are operating a systematic fund, discretion should be limited as much as possible. Portfolio decisions should follow from a clearly articulated and reproducible investment process.
  - Since we are focused on machine learning methods, quantitative methods should be at the center of your research and investment process.

  - Generally speaking, we will ignore microstructure issues such as transaction costs, slippage, and order routing. Instead, your investment process should emphasize the economic and statistical drivers of asset returns at horizons relevant for systematic investing.
  - The examples discussed in class are intended to build intuition and provide a conceptual foundation. They should not be used directly as the basis of a component of this project.
  - Your research process should distinguish between exploring a method and deploying a method. You are not expected to incorporate every method you investigate into the live portfolio.

- **Constraints**:
  - Teams should specify their complete intended portfolio exposure at all times. Portfolios may contain long and short positions and may use leverage. Teams should understand and be prepared to explain both net exposure $\sum w_i$ and gross exposure $\sum |w_i|$
  - The portfolio may contain both long and short positions and may use leverage.
  - Teams may impose additional portfolio-level constraints, but those constraints must be clearly explained and justified.
  - A Position Matrix represents the team's complete target portfolio. Unless otherwise instructed, a newly submitted Position Matrix supersedes the previously submitted target portfolio.
  - A ticker omitted from the most recent Position Matrix is interpreted as having a target weight of zero.
  - For purposes of this exercise, portfolio weights need not sum to 1 if the team uses leverage or short positions. However, the team's intended net exposure, gross exposure, and use of leverage should be clearly documented and internally consistent.

- **Performance Calculation Convention**
  - Performance Calculation Convention: For purposes of this exercise, portfolio performance will be calculated directly from submitted portfolio weights and asset returns. Daily portfolio return will be computed as $R_{p,t} = \sum w_{i,t-1} R_{i,t}$
  - Portfolio weights represent target exposures relative to portfolio value. They will not be converted into numbers of shares, and we will not maintain separate cash, margin, financing, or short-sale proceeds accounts. Unless otherwise stated, transaction costs, bid-ask spreads, slippage, short-borrow costs, and related trading microstructure issues will be ignored.

Deliverables

Each team must submit the following deliverables:

1. Position Matrices
2. Roadshow Materials
3. Due Diligence Materials

Position Matrix

- Each week, the team must submit its intended portfolio for the following investment period. You can think of this submission as the target portfolio provided to your trader.
- The Position Matrix is a CSV file with the following columns:

  Date, Ticker, Weight

- Date should take the form YYYY-MM-DD.
- Ticker should be text with no spaces; e.g., "IBM".
- Weight should be expressed in decimal form and represents the target exposure to the asset as a fraction of portfolio value. For example, a ten percent long position is 0.10 and a ten percent short position is -0.10.
- Each Position Matrix represents the team's complete target portfolio as of that submission. It is not a list of incremental trades.
- The CSV file should be titled with the submission date and team name. For example, the Jan 9, 2026 submission by the team AlphaHunters would be:

  `PositionMatrix-2026-01-09-AlphaHunters.csv`

- During the regular weekly trading cycle, the Position Matrix must be submitted by 5pm ET each Friday. Submission will take place through the team's class GitHub repository.
- If the team wishes to alter its portfolio within the week, it must submit a revised Position Matrix by 5pm ET on the trading day prior to the first trading day on which the new portfolio is to be used for performance measurement.
- For example, if on Tuesday 3Feb2026 AlphaHunters wants to alter its portfolio, the team must submit the revised Position Matrix by 5pm ET on Monday 2Feb2026:

  `PositionMatrix-2026-02-02-AlphaHunters.csv`

- The most recently submitted valid Position Matrix that has become effective determines the target portfolio used to compute official fund performance.
- These Position Matrices will be used to compute official performance reports, which will be circulated to students through the class GitHub page.
- Failure to submit a Position Matrix on time or in the required format will adversely affect the team's grade.
- Additional operational conventions, including treatment of market holidays, unavailable prices, ticker changes, or similar data issues, will be communicated by the instructor when necessary. These conventions will be applied consistently across teams.

Due Diligence Materials

In the context of our experiential activity, an institutional investor has invited your fund to provide the materials necessary to conduct a comprehensive due diligence.

The Due Diligence Materials represent the final cumulative documentation of your fund's research and investment process. They should synthesize work conducted throughout the semester rather than constitute an entirely new research project.

You are not required to introduce a new machine learning technique specifically for the Due Diligence submission. New analysis may be conducted when it is useful, but the primary objective is to organize, evaluate, document, and communicate the research your fund conducted during the semester.

You have autonomy over the organization of the deliverable, but the materials should contain the following components.

1. **Performance Summary**
   - Provide a detailed quantitative accounting of fund performance.
   - Each team has discretion regarding the specific metrics to include and the format in which they are presented.
   - At a minimum, the performance summary should include measures of return and risk on both an absolute basis and relative to the benchmark.
   - Performance should be interpreted rather than merely reported. Where appropriate, identify the major portfolio exposures, positions, or periods that contributed to fund performance.

2. **Methodology and Research Process**
   - Since you are running a quantitatively focused hedge fund, you must conduct and document a proper R&D process.
   - The discussion should include the major methods investigated during the semester, regardless of whether those methods were ultimately incorporated into the live portfolio.
   - For each major method or research question, your documentation should make clear:
     (a) What question the team was attempting to answer;
     (b) What method was used;
     (c) What evidence was generated;
     (d) What conclusion or portfolio decision followed from the evidence.
   - At a minimum, the Methodology and Research Process should include:
     - A brief primer on the major method(s) used in your R&D process. These primers should be understandable to an audience with a finance background but not necessarily a machine learning background.
     - A more detailed explanation of those methods for a technical audience.
     - The results of relevant backtests, validation exercises, diagnostics, or other empirical analysis used to evaluate proposed methods.
     - The implications of this research for your portfolio and investment process.

    - Discussion of methods that were investigated but ultimately rejected, where relevant.
  - A correctly implemented method that produces weak or unfavorable evidence may still represent high-quality research. You will not be penalized merely because a method did not improve portfolio performance, provided that the research was appropriately designed, implemented, evaluated, and interpreted.

3. **Fund Narrative**
   - Provide a written discussion of the fund's performance within the context of the market environment and the team's strategic decisions.
   - At a minimum, I would expect:
     (a) Description and interpretation of fund performance;
     (b) Rationale for outperformance or underperformance within the relevant market environment;
     (c) Discussion of important methods investigated or deployed;
     (d) Description of important strategic decisions regarding the fund's investment process or operations;
     (e) Discussion of how the fund's investment process evolved over the course of the semester.

Deliverables

- A written document created in Markdown/Quarto and rendered to HTML or PDF. You have autonomy over the layout and organization.
- The data and code necessary to reproduce the material analysis reported in the Due Diligence Materials.
- The Due Diligence submission should build upon the research repository maintained throughout the semester. Teams should not recreate previously submitted analyses solely for purposes of packaging the final deliverable.

Team Score

The score for each team's Due Diligence Materials will be allocated as follows:

- **Performance Summary (10%)**: Appropriateness and accuracy of metrics used and quality of interpretation.
- **Methodology and Research Process (50%)**: Appropriateness of methods, correctness in deployment, quality of validation, quality of interpretation, and completeness of description.
- **Code (15%)**: Code must be functional, reproducible, organized, and professionally documented.
- **Fund Narrative (15%)**: Demonstrated understanding of the relevant market environment and clear explanation of portfolio positioning, performance, and strategic decisions.
- **Professionalism (10%)**: Adherence to submission guidelines; clear organization; professional formatting and presentation.

Individual Student Scores

Your individual score for the Due Diligence Materials will be allocated as follows:

1. **Team Score (25%)**: This portion reflects the score your team received for the Due Diligence Materials.
2. **Peer Review (25%)**: You will be evaluated by your teammates. Each member receives a point budget equal to 10 points per teammate, excluding themselves. For example, in a 4-person team, each member has 30 points to allocate. Points may be distributed among teammates in any manner, including equal shares or ties, but not to oneself.

   The resulting Peer Score is given by

   $$\mathrm{PeerScore}_i = 100 \cdot \sum_{e=1}^{N-1} \frac{p_{e,i}}{\sum_j p_{e,j}},$$

   where N is the team size and $p_{e,i}$ represents the points evaluator e gave to teammate i.

   Under equal contribution, each evaluator gives equal shares to all teammates, and $\mathrm{PeerScore}_i = 100$ regardless of team size.
3. **Instructor Evaluation (50%)**: This component assesses your individual contributions across four dimensions:

   (a) understanding and application of finance concepts;

   (b) understanding and application of machine learning concepts;

   (c) quantity and quality of code;

   (d) scope and quality of contributions to team research and deliverables.

   Evaluation is informed by submitted work, attribution information, and course-related interactions inside and outside the classroom. Scores are awarded from a point budget equal to 10 points per teammate and may be allocated among team members in any manner, including equal shares or ties. Normalization will be applied as needed to ensure fairness across teams of different sizes.

Roadshows

Each team will participate in approximately five Roadshows.

Within the context of our experiential activity, Roadshows are opportunities for a hedge fund to present its investment process, research, portfolio, and performance to prospective investors for the purpose of raising capital.

You have a cumulative target capital raise of $\$100Mln$.

- The first four Roadshows focus on "friends and family" capital, with a cap of $\$5Mln$ per Roadshow.
- The fifth and final Roadshow is the Institutional Round, with a cap of $\$80Mln$.

The Roadshows will take place in class. Depending upon class size and scheduling, I may invite guests to serve in the role of prospective investors. Capital raised from these investors converts to points toward each team's Roadshow grade.

Cumulative Nature of the Roadshows

The five Roadshows should be viewed as successive versions of a single evolving fund presentation, not as five independent projects.

Between Roadshows, teams should update and improve their existing materials by:

- updating performance;
- incorporating new research;
- explaining changes to the portfolio or investment process;
- responding to evidence generated since the previous Roadshow;
- addressing weaknesses identified through instructor, investor, or peer feedback.

You are encouraged to reuse material that remains relevant. You are not expected to recreate the entire presentation, analysis, or codebase for each Roadshow.

Presentation

- We will have approximately five in-class Roadshows.
- Each team will have approximately 10–15 minutes in front of the class to present and field questions.
- To be fair to all teams, we will be strict with time allocation. When the time limit is reached, the presenting team must stop speaking.
- If a team member is answering a question when the time limit is reached, that team member may complete the response, but may not transition to another topic, nor may other team members add additional responses.

Deliverables

- PowerPoint slide deck. You have autonomy over the layout and organization. This should be the deck used during the live presentation. You may accompany the presentation with speaker notes and/or an appendix.
- Supporting data and code used in the creation of the analysis presented in the Roadshow.
- The supporting data and code should be maintained as part of your team's evolving research repository. You do not need to create an entirely separate or newly packaged codebase for each Roadshow.

Typical Structure

Although you have substantial autonomy over the organization and content of your presentation, a typical hedge fund pitch deck might contain the following elements. Not all fund styles require each element, multiple elements may appear on the same slide, and ordering may differ depending upon fund style, audience, and presentation length.

1. Title Slide
2. Legal Disclosures (you can ignore)
3. Executive Summary
4. Investment Thesis
5. Market Opportunity and Comparative Advantage
6. Strategy and Process
7. Performance
8. Risk Management
9. Team
10. Governance (you can ignore)
11. Fund Terms (you can ignore)
12. Operations and Infrastructure
13. Capital Raise
14. Appendix

For example:

- **Investment Thesis**: What economic, statistical, institutional, or behavioral mechanism do you believe creates an investment opportunity?
- **Market Opportunity and Comparative Advantage**: Why should the opportunity exist, and why is your process positioned to exploit it?

- **Strategy and Process**: How do you source ideas? How are signals constructed? How are positions sized? How do positions enter and exit the portfolio?
- **Risk Management**: What controls, limits, diversification principles, or other mechanisms influence portfolio construction?
- **Operations and Infrastructure**: You may ignore tax, legal, and compliance operations, but may discuss computing systems, data pipelines, research infrastructure, version control, or reporting processes.
- **Capital Raise**: What is your target? What are you asking investors to allocate? How does your fund justify that allocation?

Machine Learning Content

Our course will discuss the following broad classes of machine learning techniques:

- Clustering
- Dimension Reduction
- Generative Modeling
- Classification
- Regression
- Time Series

The Roadshow requirements are cumulative:

- **Roadshow 1**: At least one implementation from one of the machine learning technique families covered in the course.
- **Roadshow 2**: At least two machine learning technique families represented in the fund's research program to date.
- **Roadshow 3**: At least three machine learning technique families represented in the fund's research program to date.
- **Roadshow 4**: At least four machine learning technique families represented in the fund's research program to date.
- **Roadshow 5**: At least five machine learning technique families represented in the fund's research program to date.

These requirements refer to your fund's **cumulative research program**, not necessarily to the number of models actively controlling the live portfolio.

A machine learning technique may enter naturally into the fund's process through, for example:

- exploratory data analysis;
- signal construction;
- asset or universe segmentation;
- feature construction or dimension reduction;
- return forecasting;
- risk modeling;
- portfolio construction;
- model diagnostics;
- backtesting or validation;
- comparison of competing investment approaches.

Definition of an Implementation

For purposes of the Roadshow requirement, a machine learning technique counts as an implementation when the team has:

1. applied the technique to data relevant to the fund's research or investment process;
2. evaluated the resulting evidence using appropriate quantitative or qualitative criteria;
3. interpreted the results within the context of the fund's investment process; and
4. reached a defensible conclusion about whether and how the method should influence the fund.

Implementation does **not** imply that the team must trade directly on the output of the technique.

A valid conclusion may be to:

- deploy the method;
- modify the method;
- use the method for a supporting purpose rather than direct trading;
- retain the method for further research;
- reject the method based upon the evidence.

A correctly executed analysis that leads the team to reject a method can receive full methodological credit.

Teams have autonomy over which techniques to investigate.

Additional implementations may strengthen a team's work when they contribute meaningfully to the research process. However, quantity alone will not improve a grade.

Perfect marks can be earned with the minimum number of required implementations when those implementations are executed, evaluated, integrated, and communicated exceptionally well.

Two implementations conducted poorly are inferior to one implementation conducted well.

Team Score

The score for each team's Roadshow will be allocated as follows:

- **Instructor Evaluation (60%)**: Assesses:
  1. analytical rigor, including correctness, depth, and integration of finance and machine learning concepts;
  2. evidence and materials, including use of data, quality of analysis, clarity of visuals, and professionalism of slides;
  3. presentation effectiveness, including organization, clarity, and responsiveness to questions.
- **Audience Fundraising (10%)**: The team that raises the most capital earns 10 points toward the Roadshow grade. All other teams' scores are prorated relative to that benchmark. For example, if Fund B raises 90% of the capital raised by Fund A, then Fund B earns

$$0.9 \times 10 = 9$$

  points.
- **Supporting Data and Code (30%)**: Evaluation of the analysis supporting the Roadshow, including functionality, reproducibility, organization, documentation, and the quality of the underlying research process.

Audience fundraising is intended to reward the credibility and effectiveness of your investment presentation. Investors should consider the quality of the investment process, evidence, risk management, communication, and realized performance rather than simply allocating capital to the fund with the highest trailing return.

If we do not have guest investors, the point allocation will be:

- Instructor Evaluation: 70%
- Supporting Data and Code: 30%

Individual Student Scores

Your individual score for the Roadshows will be allocated as follows:

1. **Team Score (25%)**: This portion reflects the score your team received for the Roadshows.
2. **Peer Review (25%)**: You will be evaluated by your teammates. Each member receives a point budget equal to 10 points per teammate, excluding themselves. For example, in a 4-person team, each member has 30 points to allocate. Points may be distributed among teammates in any manner, including equal shares or ties, but not to oneself.

The resulting Peer Score is given by

$$\text{PeerScore}_i = 100 \cdot \sum_{e=1}^{N-1} \frac{p_{e,i}}{\sum_j p_{e,j}},$$

where N is the team size and $p_{e,i}$ represents the points evaluator e gave to teammate i.

Under equal contribution, each evaluator gives equal shares to all teammates, and $\text{PeerScore}_i = 100$ regardless of team size.

3. **Instructor Evaluation (50%)**: This component assesses individual contributions across four dimensions:

   (a) understanding and application of finance concepts;

   (b) understanding and application of machine learning concepts;

   (c) quantity and quality of code;

   (d) scope and quality of contributions to team research and deliverables.

   Evaluation is informed by submitted work, attribution information, and course-related interactions inside and outside the classroom. Scores are awarded from a point budget equal to 10 points per teammate and may be allocated among team members in any manner, including equal shares or ties. Normalization will be applied as needed to ensure fairness across teams of different sizes.

Graduate Research Extension

Students enrolled in Econ690 must complete an individual Graduate Research Extension related to their team's hedge fund.

The purpose of the Graduate Research Extension is to provide master's students an opportunity to demonstrate additional analytical depth, methodological judgment, and independence within the context of the team's broader research program.

The Graduate Research Extension should arise naturally from a research question relevant to the fund. It is not intended to be an unrelated research project, nor does it require the use of an additional machine learning technique. The emphasis is on depth of analysis rather than quantity of analysis.

The extension may involve, for example:

- additional model validation or robustness analysis;
- comparison of alternative model specifications or methodological choices;
- investigation of model stability or sensitivity to assumptions;
- analysis of estimation error or uncertainty;
- alternative approaches to portfolio construction or risk management;
- investigation of data quality, look-ahead bias, leakage, or other empirical design issues;
- analysis of whether statistical improvements translate into economically meaningful portfolio outcomes;
- another substantive extension approved by the instructor.

Research Process

The Graduate Research Extension should clearly document the following:

1. **Question**: What specific issue is being investigated, and why does it matter for the team's fund?
2. **Method**: What additional analysis was performed, and why is the method appropriate for answering the research question?
3. **Evidence**: What results, diagnostics, robustness checks, model comparisons, or other evidence were generated?
4. **Conclusion**: What did the analysis show, and what implication, if any, should the evidence have for the fund's investment process?

As with the broader hedge fund exercise, a Graduate Research Extension does not need to generate favorable results or improve realized portfolio performance. A carefully designed and correctly implemented analysis that leads to the conclusion that the fund should not alter its process may receive full credit.

Integration with the Team Project

- The Graduate Research Extension is an individual requirement and should represent a substantive contribution by the Econ690 student.
- The research should be related to the student's team fund and should use or extend the team's data, models, investment process, or research infrastructure where appropriate.
- The analysis and supporting code should be maintained within the team's research repository. The graduate student's contribution must be clearly attributable.
- Evidence generated through the Graduate Research Extension may be incorporated into the team's Roadshows, investment process, and Due Diligence Materials when relevant.
- Other members of the team are not responsible for completing the Graduate Research Extension.

Research Proposal

Each Econ690 student must submit a brief proposal describing the intended research question and proposed analysis. The proposal must be approved by the instructor before the student proceeds with the extension. See Canvas for the proposal due date and submission instructions.

Evaluation

The Graduate Research Extension will be evaluated based upon:

- **Research Question**: Importance, clarity, and relevance of the question to the team's fund.
- **Methodological Rigor**: Appropriateness of the empirical or quantitative approach and the quality of methodological judgment demonstrated.
- **Implementation and Evidence**: Correctness and reproducibility of the analysis and the quality of the evidence generated.
- **Interpretation**: Depth and accuracy of interpretation, including recognition of relevant limitations and uncertainty.
- **Investment Relevance**: Quality of the connection between the evidence and the fund's investment or research process.
- **Independence and Professionalism**: Evidence of substantive individual contribution, clear documentation, and professional presentation of the work.

Research Repository and Submission Instructions

Your GitHub repository should function as a **living research repository** for your fund throughout the semester.

The purpose of the repository is to allow your team and the instructor to follow the evolution of your research process and reproduce the analysis underlying your investment decisions and deliverables.

You are not expected to create a separate, newly packaged codebase for every Roadshow. Instead, your team's data, code, analysis, and documentation should evolve over time as the fund develops.

- Once teams are formed, we will create a GitHub repository for each team and invite each student to be a collaborator on the team repository.
- We will also create groups on Canvas organized by team in order to store grades and facilitate communication.
- Your team repository should contain clearly organized directories for:
  - Data
  - Code
  - Deliverables
- Within the Deliverables directory, folders should be named according to each deliverable. For example:
  - `Roadshow1`
  - `Roadshow2`
  - `Roadshow3`
  - `Roadshow4`
  - `Roadshow5`
  - `DueDiligence`
- Roadshow folders should contain the presentation materials associated with that Roadshow. Supporting analysis may remain within the shared Data and Code directories rather than being duplicated for every presentation.
- All materials relevant to a graded deliverable must be committed to the repository by the stated due date and time.
- Please include appropriate README documentation within the repository so that another technically competent person can understand the organization of the project and reproduce the relevant analysis.

Research Documentation

Your repository should provide a clear record of the team's major research decisions.

For each substantial research experiment, the documentation should make it possible to identify:

1. **Question**: What was the team trying to learn?

2. **Method**: What analysis or machine learning technique was used?
3. **Evidence**: What did the analysis show?
4. **Decision**: What implication did the evidence have for the portfolio or investment process?

This documentation does not need to take the form of a separate graded assignment. It may appear naturally within notebooks, scripts, Quarto documents, README files, or other project documentation.

The purpose is to preserve the reasoning behind the team's research decisions so that the final Due Diligence Materials can synthesize work completed throughout the semester rather than reconstruct the research process after the fact.

Attribution

Provide clear attribution of team members' contributions.

- For slide decks, attribution might appear as footnotes on individual slides, attribution by section, or a summary at the end of the deliverable.
- For code, attribution might appear through author information, comments associated with major sections, commit history, or other documentation that clearly identifies meaningful contributions.
- Attribution should make it possible to distinguish the substantive contributions of individual team members without requiring unnecessary administrative work.

File Naming

- **Due Diligence Materials**: The primary written deliverable should be rendered to HTML or PDF. If submitted as a PDF, use the following naming convention:

  `DueDiligence-TeamName.pdf`

  For example:

  `DueDiligence-ABC.pdf`

- **Roadshows**: The file type should be PPTX with the following naming convention:

  `Roadshow-TeamName-Number.pptx`

  For example, the first Roadshow for team ABC should be named:

  `Roadshow-ABC-1.pptx`